



Telegraph

A peer-to-peer ranking protocol for machine intelligence.

telegraphprotocol.com

Telegraph in full.

What it is

Telegraph ranks machine intelligence and sells the winning answer. Any model, dataset, tool or API plugs in as a miner with one config file, and each request is routed to whoever ranks best, from \$0.01.

How the ranking works

Ranking runs continuously in the background, per intent. 64 independent validators score miners by running that intent's Evaluator in a sandbox, with 43 of 64 needed to finalise.

Why it stays honest

No single company runs the scoring, so nobody can buy the top spot. The tests compete too: a better Evaluator takes over scoring. Fraud forfeits rewards. Downtime ejects.

Who does what

Miners supply the intelligence. Evaluators write the scoring logic. Validators run it and agree the ranking. Agents pay for answers. The protocol builds none of it and takes 2%.

How money moves

Agents pay per request in USDC on Ethereum. The protocol keeps 2% and converts 98% into Machina via TWAP to pay the winning miner, so every paid request is a market buy.

The token and the proof

Machina: 21M hard cap, four-year halvings, zero pre-mine, zero team or VC allocation, 60/20/20 emissions. Season I drew 1,400+ miners, 120K testnet transactions, and 45 apps. Mainnet Dec 26 / Jan 27.

Fixed, Bitcoin-style, demand-linked

- 21,000,000 Machina hard cap; 7,200 per 24-hour emission period at genesis; halving every 4 years; zero pre-mine, zero team / VC allocation.
- Emission split (immutable): Validators 60% / Evaluators 20% / Treasury 20%.
- Value linkage - every paid request from a human, app, or agent converts their USDC into Machina via continuous TWAP, so real usage is a mechanical market buy, not speculation.
- Immutable forever: cap, halving, 60/20/20 split, zero pre-mine; other parameters need 43-of-64 plus a timelock.

21M

hard cap, fixed

60 / 20 / 20

validator / evaluator / treasury

0%

miner emissions

Why operating a node is a major opportunity

A node puts you inside the agent economy - the rail every paid machine request for the top-ranked answer runs through.

- Own a position in the agent economy - one of 64 nodes ranking what agents pay to use.
- Sit on the ranking - every paid request routes using ranking state finalized by the validator set.
- Earn from several streams - emissions, a USDC fee share, staking commission, and even your own miners.

Run a node. Earn two ways.

The network is ranked and secured by 64 validator nodes. As an operator, your node earns from two independent streams every day the network runs. Every active node earns the same emission share whether it ranks 1st or 64th - no need to become a whale to earn well, and no politics.

STREAM ONE

Protocol emissions

- 60% of 7,200 = 4,320 Machina/day, split equally across 64 nodes = 67.5 Machina per node per day.
- Approx. 24,637 Machina per node per year before the first halving - equal whether ranked 1st or 64th.
- Plus redistributed emissions, testing-cohort fee, Evaluator rewards, and delegated-staking commission.

STREAM TWO

USDC fee share (Roadmap)

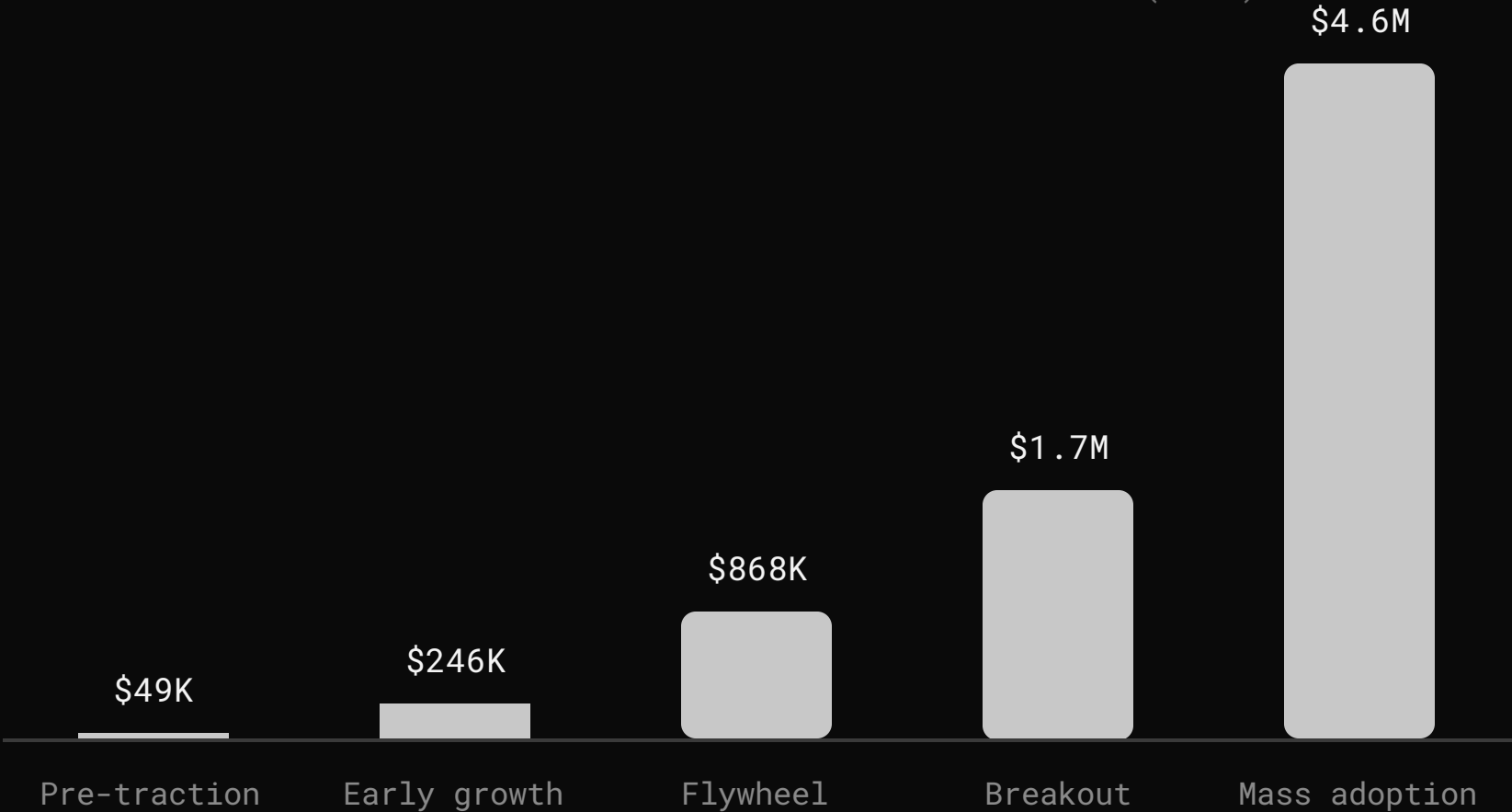
- The protocol currently collects a 2% fee on every paid request, in USDC, directed to the Treasury.
- A future mechanism is planned to allocate up to 1% of this fee to active validators as USDC, a non-volatile revenue stream alongside Machina emissions.
- Not active at genesis; introduced at a later stage based on network growth and validator participation.



What a single node could earn as the network grows

Earnings per node climb with request volume. These figures are transaction-fee income, not the emission rewards nodes also receive. The chart shows the high end of each stage; the table shows the full modelled range.

SINGLE-NODE ANNUAL EARNINGS - HIGH END OF RANGE (USD)



Network stage	Daily requests	Per node / yr
Pre-traction	~500	\$16K - \$49K
Early growth	~5,000	\$81K - \$246K
Flywheel active	~50,000	\$246K - \$868K
Breakout	~500,000	\$578K - \$1.66M
Mass adoption	2,000,000+	\$1.4M - \$4.6M

When intelligence gets cheaper, being right becomes more valuable.

Inference keeps getting cheaper, so the model stops being the moat. Knowing which answer to trust becomes the scarce part. No frontier lab can subsidise supply the way an open market competing for rewards does. Telegraph ranks that competition, verifies the winner neutrally, and sells the answer.

BUILD IT YOURSELF

\$78M – \$191M

to train one frontier model (GPT-4 to Gemini Ultra)

- AI / ML engineers at \$160K-\$300K each, per year
- \$500K+ for a single 8-GPU server, plus power and ops
- 80%+ of AI projects fail to reach value (RAND)
- You still build verification and monitoring yourself

BUY THE ANSWER

\$0.01

per request - the top-ranked answer from the entire market

- No training, no GPUs, no ML team to hire
- Verified by 64 neutral validators, not by a vendor
- Routed to whoever ranks top for that intent
- You buy answers per intent, not model seats



How we attract supply and dominate the agentic market.

Bitcoin-style money for machines: fixed supply, pointed at intelligence instead of hashing. No lab paying fiat can subsidise supply the way open token rewards do, and mining needs no rig - just an API, model or dataset.

	BITCOIN	BITTENSOR (\$TAO)	TELEGRAPH
What you do	Solve cryptographic puzzles.	Run models / provide intelligence in subnets.	Supply any intelligence behind an API (model, data feed, tool, etc.).
Hardware	ASICs (\$4,000 - \$10,000+ per machine).	High-end GPUs or dedicated cloud compute clusters.	Often none. Many miners just use an existing API + a YAML config file.
Ongoing cost	High electricity and cooling (~\$150 - \$600+ per month per machine; ~\$32,000 to \$59,000 in power per BTC mined).	Dynamic subnet slot registration fee (burned in TAO; typically \$200 - \$2,000+) + monthly GPU hosting (\$300 - \$2,000+/month).	Near zero. Main cost is whatever is already paid to run the existing API (\$0 - \$10/month serverless overhead).
Barrier	Very high.	High.	Extremely low.
Paid in	Block rewards + fees (emissions).	Subnet emissions (TAO).	Real demand only. Agents pay USDC, the protocol converts to Machina via TWAP and sends it to the winning miner. Zero emissions.
Total cost to participate	\$4,000 - \$10,000+ upfront per ASIC + \$150 - \$600+/month recurring.	\$500 - \$2,000+ upfront + \$300 - \$2,000+/month recurring compute.	~\$0 upfront + ~\$0 - \$100/month, depending on the kind of intelligence you provide.

Keys, provenance, consensus, and penalties

Key separation

Manager Key (cold) holds stake offline; Operator Key (hot) only signs. A breach can't touch principal; keys rotate on-chain.

Provenance (zkTLS)

zkTLS binds the Miner endpoint to the exact input and response; invalid or substituted evidence is rejected.

Consensus (BFT)

43 of 64 must sign to finalise (tolerates 21 bad/offline); stake-weighted median, commit-reveal, BLS-aggregated.

Credentials

Miner API keys encrypted across hardware enclaves; no single node holds the full key; auto-rotated on secure hardware.

Penalties

Liveness = ejection; fraud = forfeiture + permanent ban; incorrect execution = forfeiture; principal never slashed.

Abuse resistance

Economic Sybil cost on a fixed 21M supply; jittered TWAP settlement makes front-running unprofitable.

What it costs and how to deploy

PRE-MAINNET (CURRENT PHASE)

Running a node during testnet is extremely lightweight.

- Infrastructure - a standard server with 8-16 GB RAM; no GPUs or specialized hardware required.
- Cost - low hundreds of USD per month, plus a couple of testnet tokens.
- Setup - run a single script and bind a private key; no staking or capital at this stage.
- Staff - minimal oversight (uptime and key management).

Run a node immediately with almost zero financial commitment or operational complexity.

POST-MAINNET

Requirements shift to operating as a full validator.

- Capital - stake enough Machina to remain in the top 64 (market-determined, 21-day unbonding).
- Infrastructure - still lightweight, no GPUs; a secure enclave-capable server; submission gas reimbursed.
- Setup - stake via Manager Key (cold), bind an Operator Key (hot), deploy one config file, sync to NTP.
- Staff - minimal; uptime monitoring and basic key hygiene.

Compound further through delegated staking, running your own miners, and onboarding ecosystem agents.



What a node costs to run

Pre-mainnet, a node needs 4 vCPU, 8 GB RAM, 30-50 GB SSD, Ubuntu 22.04 with Docker and port 7044 open. At mainnet, infrastructure costs are real but scale far more slowly than rewards.

NETWORK STAGE	DAILY REQUESTS	ANNUAL EARNINGS	MONTHLY COST	NET ANNUAL YIELD
Pre-traction	~500	\$16K - \$49K	~\$200 - \$400	~\$48K
Early growth	~5,000	\$81K - \$246K	~\$300 - \$500	~\$245K
Flywheel active	~50,000	\$246K - \$868K	~\$400 - \$700	~\$867K
Breakout	~500,000	\$578K - \$1.66M	~\$500 - \$900	~\$1.65M
Mass adoption	2M+	\$1.4M - \$4.6M	~\$800 - \$1,500	~\$4.58M

- Validators evaluate intelligence, they do not generate it - stateless WASM scoring scripts run in milliseconds on standard CPUs, with no GPU clusters to fund.
- At block production and epoch transitions the node runs the full registry of WASM binaries across multiple cores - thousands of scripts in under a second.
- Costs scale predictably with volume. Even at mass adoption, an enterprise cloud budget of up to \$1,500 a month is a small fraction of a node's projected earnings.



Σ Telegraph

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